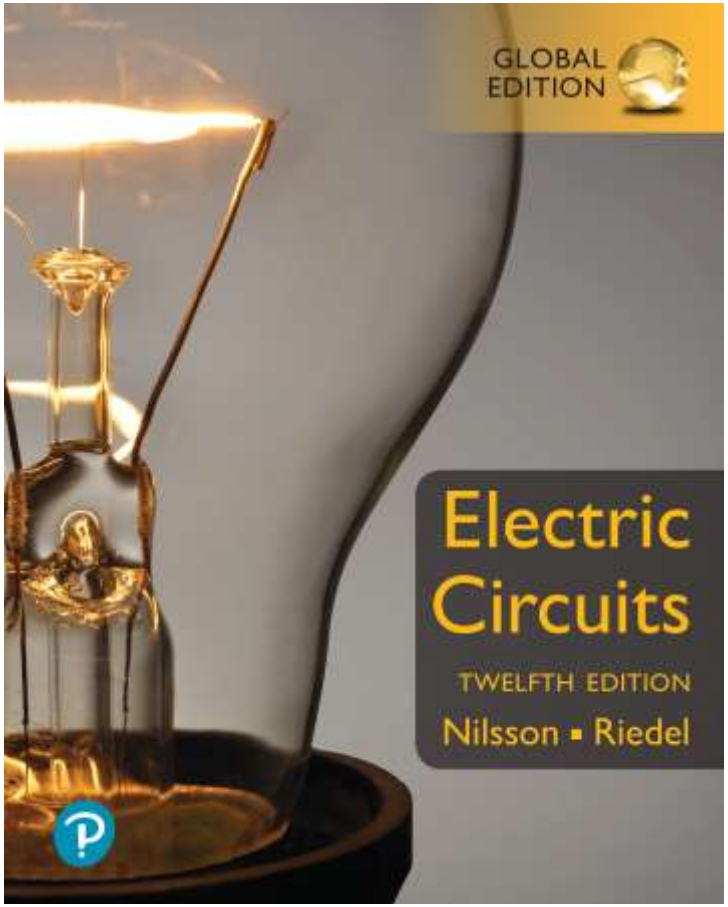
# Electric Circuits

Twelfth Edition, Global Edition




## Chapter 7 – Part 4

Response of First-Order *RL* and *RC* Circuits

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# Chapter 7 – Sequential Switching Problems (1 of 22)

Sometimes first-order RL/RC circuits have multiple switching times. This can happen when a circuit has

- One switch that goes back and forth between two positions at different times
- One switch that has three or more positions and at least two different switching times
- Two or more switches whose positions change at different times.

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# Chapter 7 – Sequential Switching Problems (2 of 22)

- Such circuits can exhibit the natural response, the step response, or both!
- We analyze the behavior of a sequential switching circuit using the same analysis method used for first-order circuits with only one switching time, repeating the process for each time segment!
- The next slide reviews the analysis method. Examples will be used to illustrate.

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# Chapter 7 – *RL*/*RC* Natural and Step Response (1 of 2)

This is a general solution method for both the natural and step responses of first-order circuits. A circuit exhibiting a natural response has no independent source as $t \to \infty$, so its final value $x_f = 0$.

- Step 1 – Identify the variable $x(t)$, which is the inductor current for *RL* circuits and capacitor voltage for *RC* circuits
- Step 2 – Calculate the initial value $x_0$ by analyzing the circuit to find $x(t)$ for $t < 0$.
- Step 3 – Calculate the time constant $\tau$; for *RL* circuits, $\tau = R/L$ and for *RC* circuits, $\tau = RC$, where $R$ is the equivalent resistance connected to the inductor or capacitor for $t \geq 0$.
- Step 4 – Calculate the final value $x_f$ by analyzing the circuit to find $x(t)$ as $t \to \infty$.

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# Chapter 7 – R L/R C Natural and Step Response (2 of 2)

- Step 5 – Write the equation for $x(t)$,

$$x(t) = X_f + (X_0 - X_f)e^{-t/\tau}, \text{ for } t \geq 0.$$

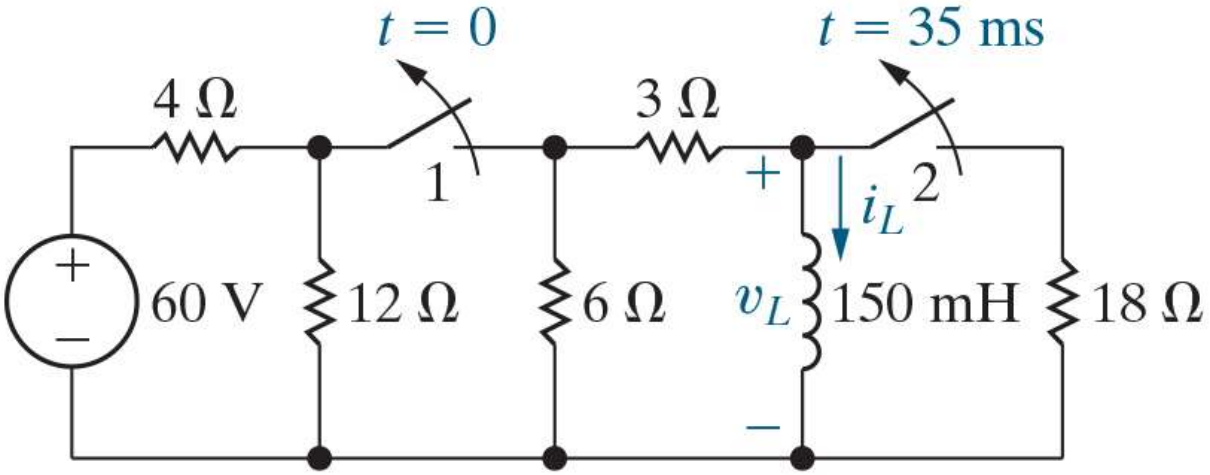
- Step 6 – Calculate other quantities of interest using $x(t)$.

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# Chapter 7 – Sequential Switching Problems (3 of 22)

Both switches have been closed for a long time. At t = 0, switch 1 is opened and stays open. Switch 2 opens 35 ms later and stays open. Find the expression for the voltage drop across the inductor for $t \geq 0^+$.

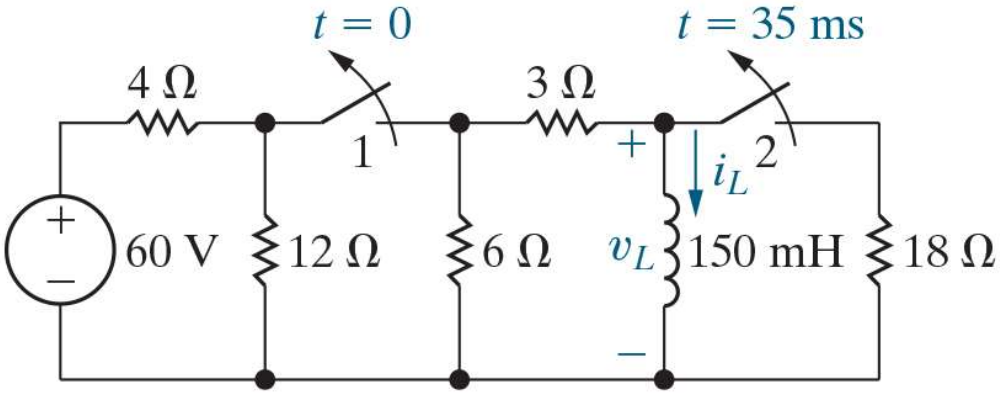



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# Chapter 7 – Sequential Switching Problems (4 of 22)

How many different time segments are defined for this circuit for $t \geq 0^+$?




A. 1

B. 2

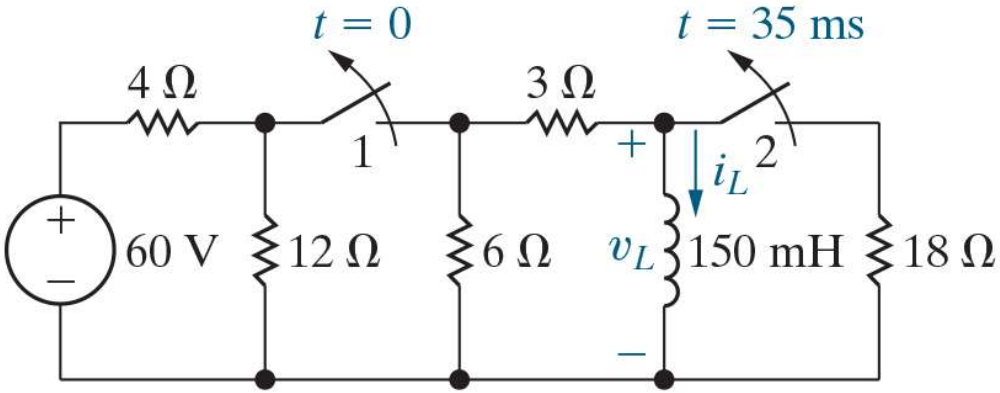
C. 3

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# Chapter 7 – Sequential Switching Problems (4 of 22)

How many different time segments are defined for this circuit for $t \geq 0^+$?




✗ A. 1

✓ B. 2

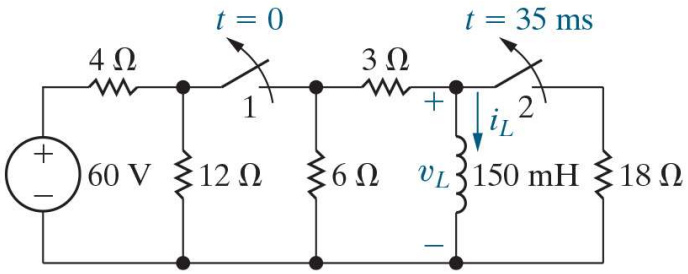
✗ C. 3

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# Chapter 7 – Sequential Switching Problems (5 of 22)

Both switches have been closed for a long time. At t = 0, switch 1 is opened and stays open. Switch 2 opens 35 ms later and stays open. Find the expression for the voltage drop across the inductor for $t \geq 0^+$.



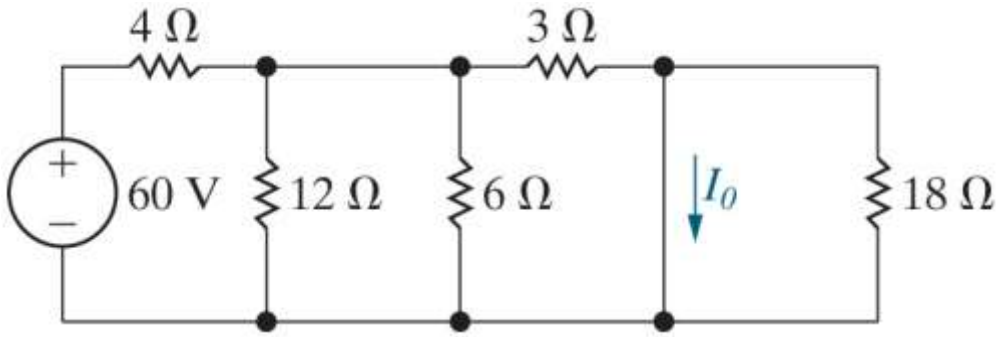

- Analysis for the first time segment, $0 \leq t \leq 35$ ms:
  - Step 1 – Identify the variable $x(t)$, which is the inductor current $i_L$.
  - Step 2 – Calculate the initial value $I_0$ by analyzing the circuit to find $i_L(t)$ for $t < 0$.

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# Chapter 7 – Sequential Switching Problems (6 of 22)

Here is the circuit for $t < 0$. The current $I_0$ is the current in which resistor?




A. 3 Ω

B. 18 Ω

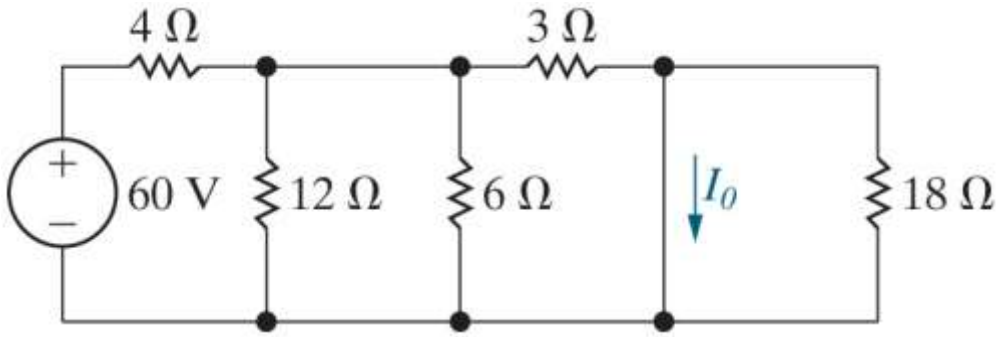
C. 4 Ω

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# Chapter 7 – Sequential Switching Problems (6 of 22)

Here is the circuit for $t < 0$. The current $I_0$ is the current in which resistor?




✔ A. 3 Ω

✘ B. 18 Ω

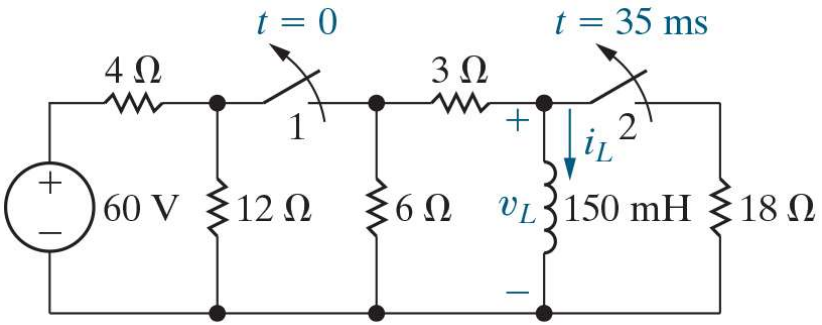
✘ C. 4 Ω

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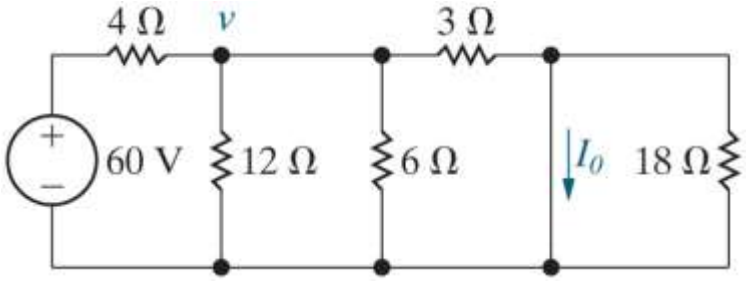
# Chapter 7 – Sequential Switching Problems (7 of 22)

Both switches have been closed for a long time. At t = 0, switch 1 is opened and stays open. Switch 2 opens 35 ms later and stays open. Find the expression for the voltage drop across the inductor for $t \geq 0^+$.




- Analysis for the first time segment, $0 \leq t \leq 35$ ms:
  - Step 2 – Calculate the initial value $I_0$ by analyzing the circuit to find $i_L(t)$ for $t < 0$. Write a KVL equation.

t < 0:




$$\frac{v-60}{4}+\frac{v}{12}+\frac{v}{6}+\frac{v}{3}=0$$

$$\Rightarrow \quad v = 18 \text{ V}$$

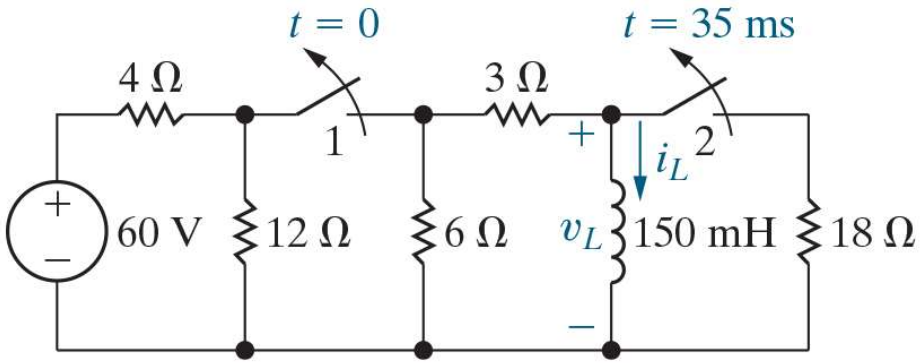
$$\Rightarrow \quad I_0 = 18/3 = 6 \text{ A}$$

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# Chapter 7 – Sequential Switching Problems (8 of 22)

Next we need to find the final value of the inductor current, $i_L(\infty) = I_F$. To do this, analyze the circuit with the switches in the following positions:



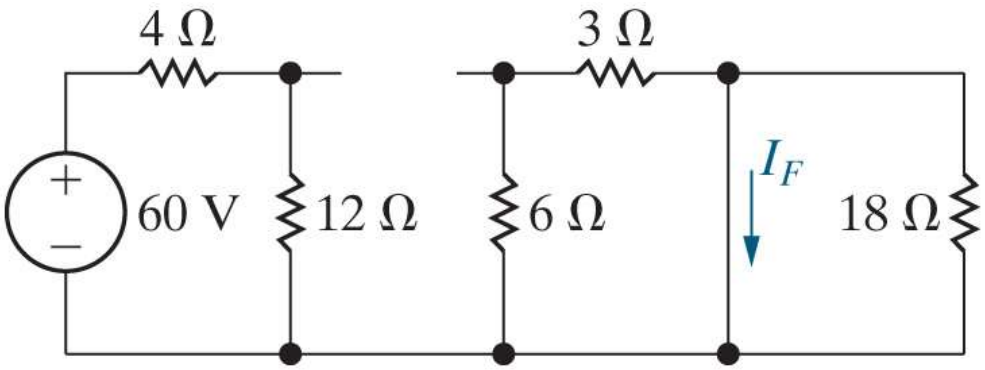

- ✗ A. SW 1 closed, SW 2 closed
- ✓ B. SW 1 open, SW 2 closed
- ✗ C. SW 1 open, SW 2 open

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# Chapter 7 – Sequential Switching Problems (9 of 22)

Here is the circuit used to determine the final value of the current in the first time segment. For this time segment, is the response form the natural response or the step response?




A. Natural response

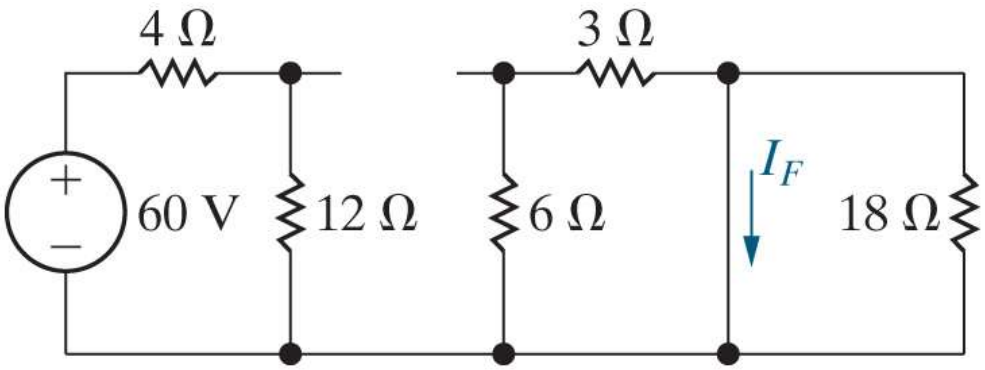
B. Step response

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# Chapter 7 – Sequential Switching Problems (9 of 22)

Here is the circuit used to determine the final value of the current in the first time segment. For this time segment, is the response form the natural response or the step response?




✔ A. Natural response

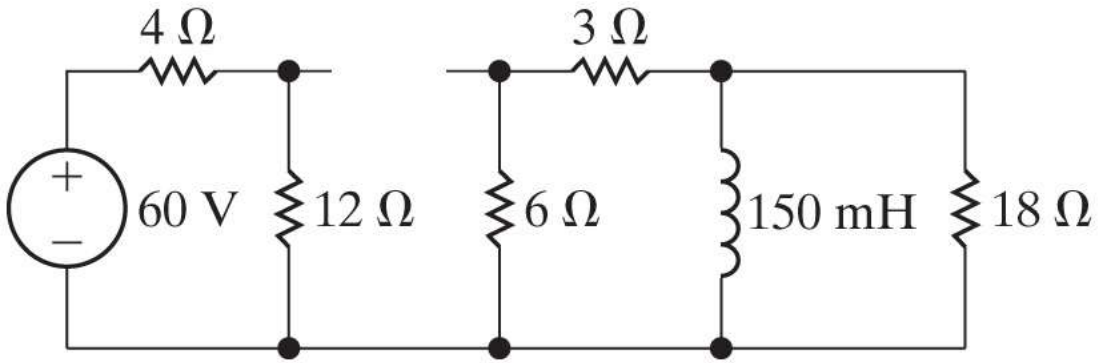
✘ B. Step response

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# Chapter 7 – Sequential Switching Problems (10 of 22)

Next, calculate the time constant for this time segment. To do this, find the equivalent resistance seen by the inductor for the circuit that exists in this time segment.




A. $(6+3+18)\ \Omega$

B. $[(4\,||\,12\,||\,6)+3]\ \Omega$

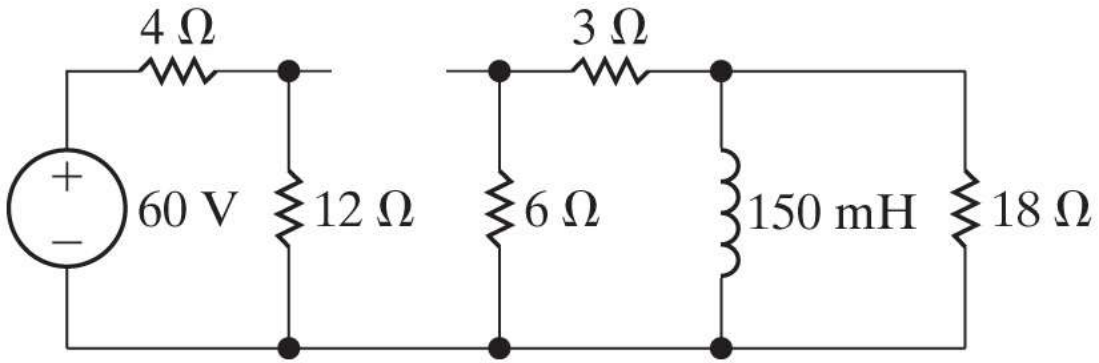
C. $[(6+3)\,||\,18]\ \Omega$

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# Chapter 7 – Sequential Switching Problems (10 of 22)

Next, calculate the time constant for this time segment. To do this, find the equivalent resistance seen by the inductor for the circuit that exists in this time segment.




✗ A. $(6+3+18)\ \Omega$

✗ B. $[(4\,||\,12\,||\,6)+3]\ \Omega$

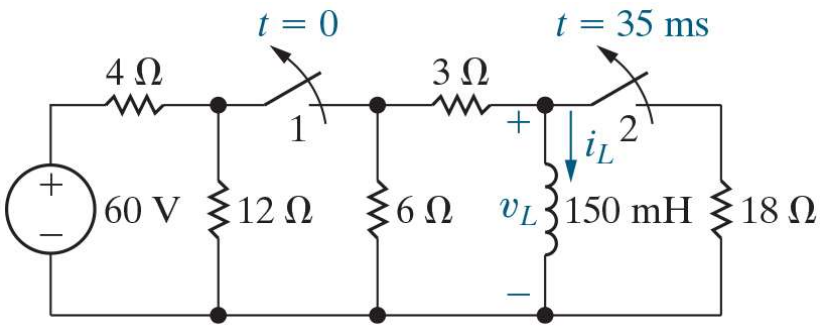
✓ C. $[(6+3)\,||\,18]\ \Omega$

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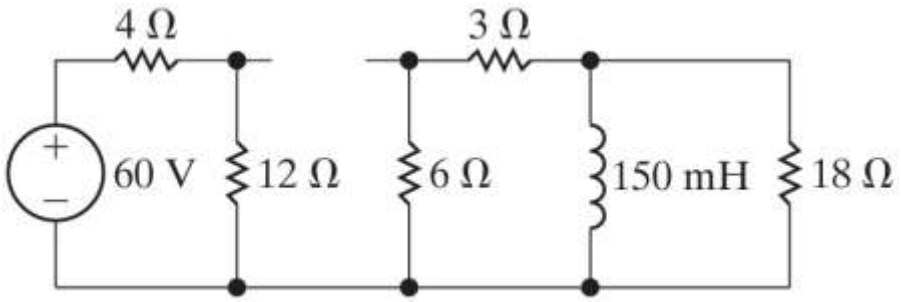
# Chapter 7 – Sequential Switching Problems (11 of 22)

Both switches have been closed for a long time. At t = 0, switch 1 is opened and stays open. Switch 2 opens 35 ms later and stays open. Find the expression for the voltage drop across the inductor for $t \geq 0^+$.




- Analysis for the first time segment, $0 \leq t \leq 35$ ms:
  - Step 3 – Calculate the time constant $\tau = L/R$.

t < 0:




$$R_{eq} = (6+3) \| 18$$

$$= 9 \| 18 = 6\Omega$$

$$\tau = L/R_{eq} = 0.15/6$$

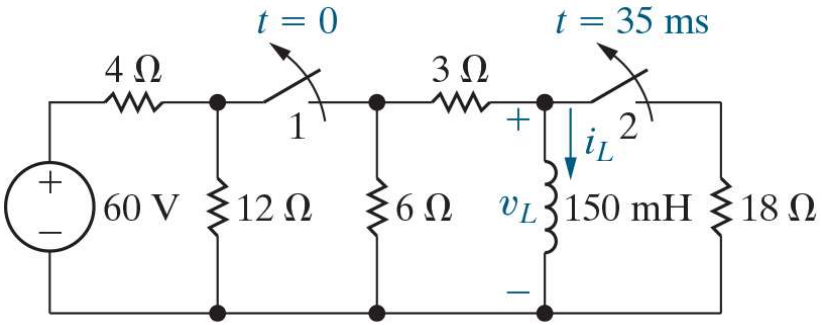
$$= 0.025 \text{ s}$$

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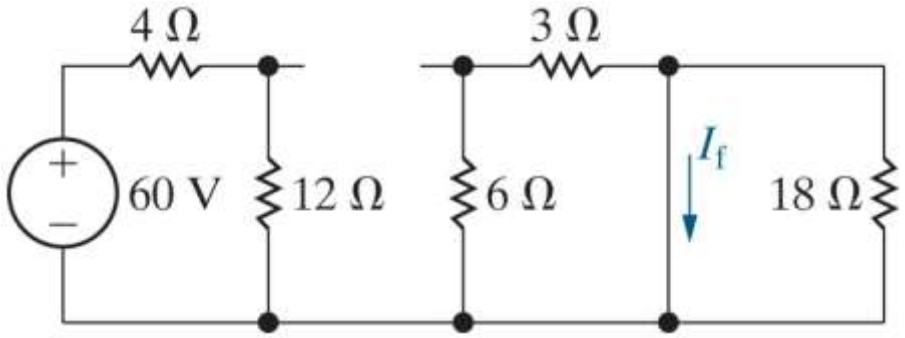
# Chapter 7 – Sequential Switching Problems (12 of 22)

Both switches have been closed for a long time. At t = 0, switch 1 is opened and stays open. Switch 2 opens 35 ms later and stays open. Find the expression for the voltage drop across the inductor for $t \geq 0^+$.




- Analysis for the first time segment, $0 \leq t \leq 35$ ms:
  - Step 4 – Calculate the final value $I_f$.

$t \to \infty$:




Natural response!

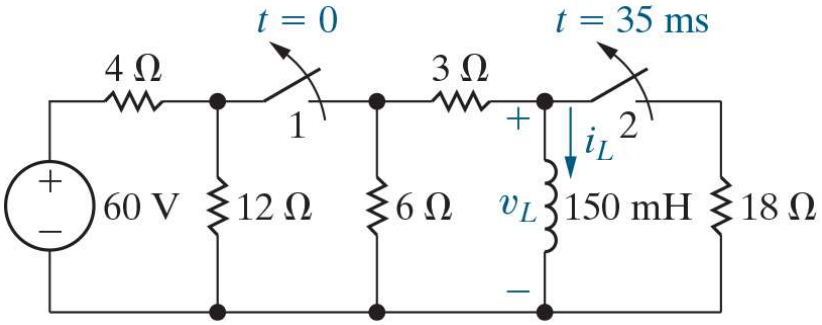
$\Rightarrow \quad I_f = 0$ A

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# Chapter 7 – Sequential Switching Problems (13 of 22)

Both switches have been closed for a long time. At t = 0, switch 1 is opened and stays open. Switch 2 opens 35 ms later and stays open. Find the expression for the voltage drop across the inductor for $t \geq 0^+$.




- Analysis for the first time segment, $0 \leq t \leq 35$ ms:
  - Step 5 – Write the equation for $i_L(t)$.

$$i_L(t) = I_f + (I_0 - I_f)e^{-t/\tau}$$

$$= 0 + (6 - 0)e^{-t/0.025}$$

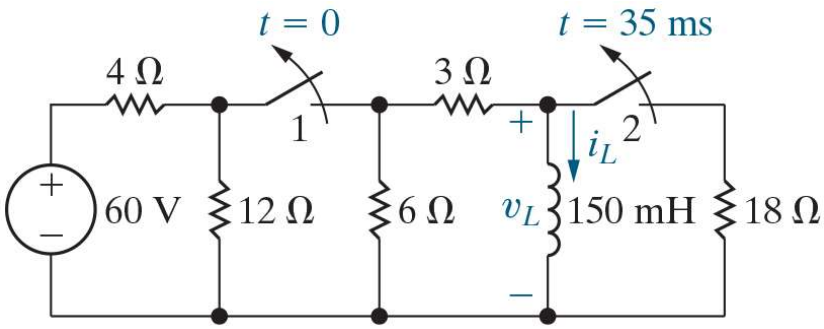
$$= 6e^{-40t} \text{ A}, \quad 0 \leq t \leq 35 \text{ ms}$$

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# Chapter 7 – Sequential Switching Problems (14 of 22)

Both switches have been closed for a long time. At t = 0, switch 1 is opened and stays open. Switch 2 opens 35 ms later and stays open. Find the expression for the voltage drop across the inductor for $t \geq 0^+$.




- Analysis for the second time segment, $t \geq 35$ ms:
  - Step 2 – Calculate the initial value $I_0$. Since the inductor current must be continuous for all time, this is the value of the inductor current at the end of the last time segment.

$$I_0 = i_L(35 \text{ ms})$$

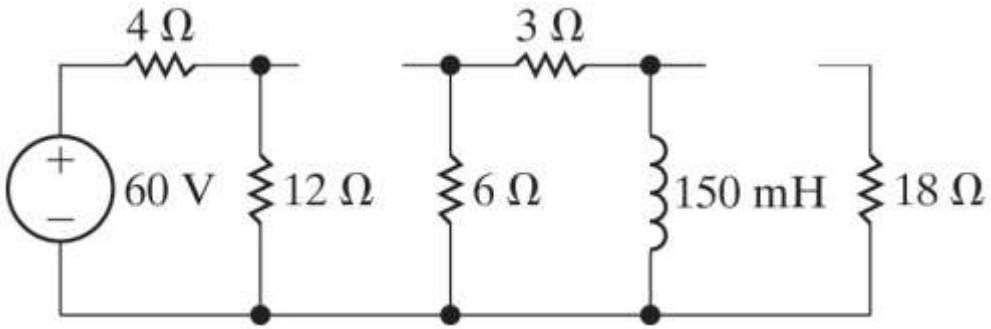
$$= 6e^{-40(0.035)} = 1.48 \text{ A}$$

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# Chapter 7 – Sequential Switching Problems (15 of 22)

Next, calculate the time constant for this time segment. To do this, find the equivalent resistance seen by the inductor for the circuit that exists in this time segment.




A. $(6+3+18)\ \Omega$

B. $(6+3)\Omega$

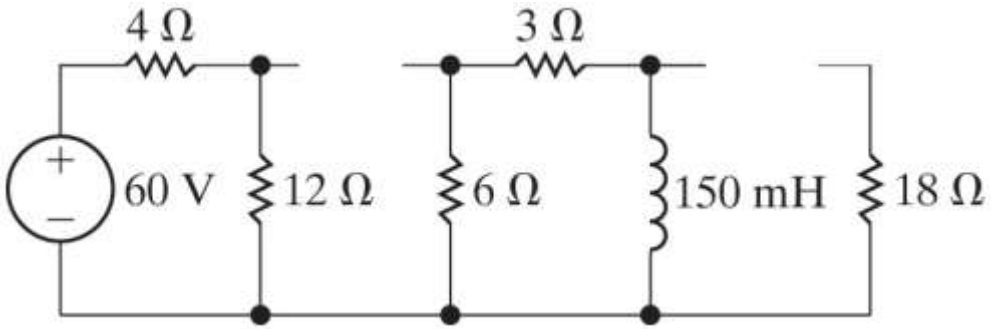
C. $[(6+3)\,||\,18]\ \Omega$

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# Chapter 7 – Sequential Switching Problems (15 of 22)

Next, calculate the time constant for this time segment. To do this, find the equivalent resistance seen by the inductor for the circuit that exists in this time segment.




✗ A. $(6+3+18)\ \Omega$

✓ B. $(6+3)\Omega$

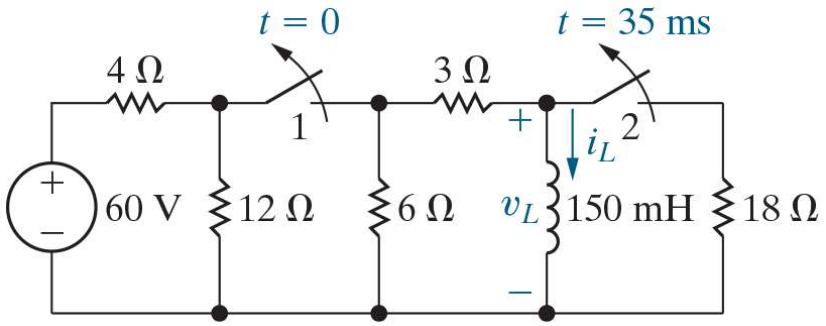
✗ C. $[(6+3)\,||\,18]\ \Omega$

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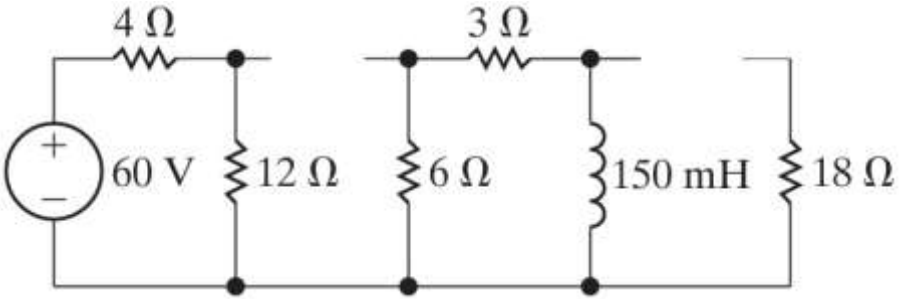
# Chapter 7 – Sequential Switching Problems (16 of 22)

Both switches have been closed for a long time. At t = 0, switch 1 is opened and stays open. Switch 2 opens 35 ms later and stays open. Find the expression for the voltage drop across the inductor for $t \geq 0^+$.




- Analysis for the second time segment, $t \geq 35$ ms:
  - Step 3 – Calculate the time constant $\tau = L/R$.

$t \geq 35$ ms:




$$R_{eq} = (6+3) = 9\Omega$$

$$\tau = L/R_{eq} = 0.15/9$$

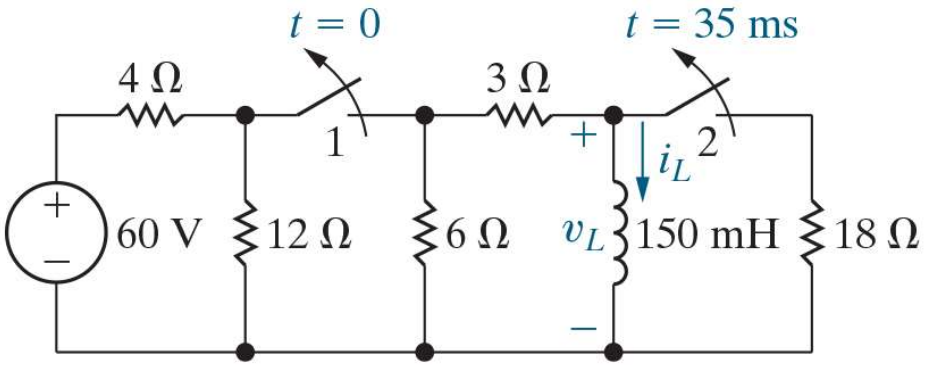
$$= 0.0167 \text{ s}$$

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# Chapter 7 – Sequential Switching Problems (17 of 22)

Next we need to find the final value of the inductor current, $i_L(\infty) = I_F$ for the second time segment. To do this, analyze the circuit with the switches in the following positions:




A. SW 1 closed, SW 2 closed

B. SW 1 open, SW 2 closed

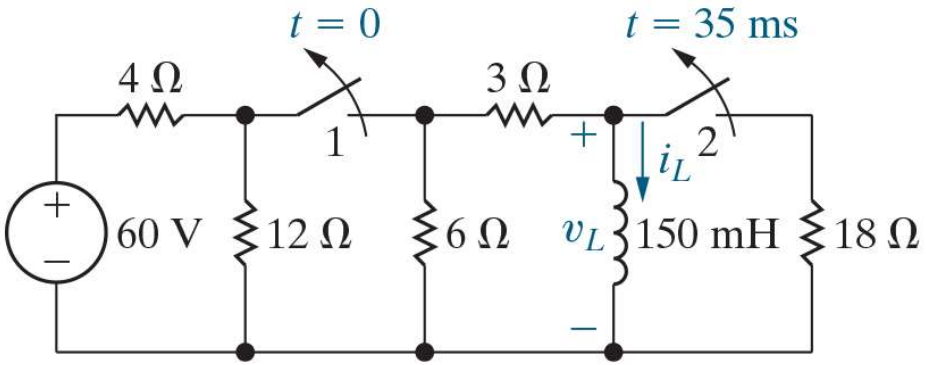
C. SW 1 open, SW 2 open

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# Chapter 7 – Sequential Switching Problems (17 of 22)

Next we need to find the final value of the inductor current, $i_L(\infty) = I_F$ for the second time segment. To do this, analyze the circuit with the switches in the following positions:



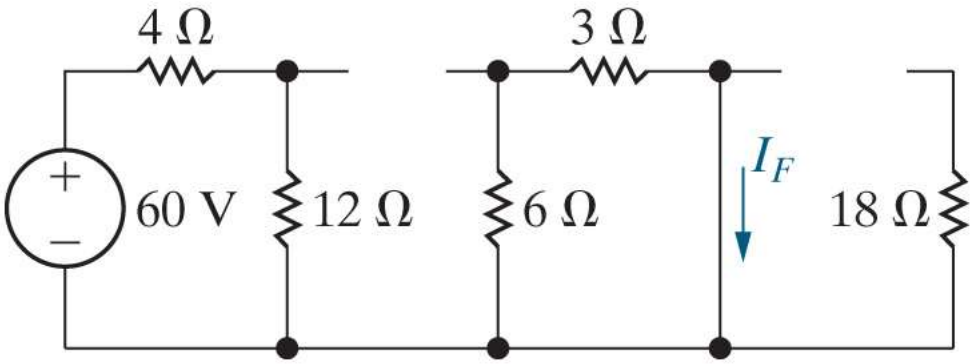

- ✗ A. SW 1 closed, SW 2 closed
- ✗ B. SW 1 open, SW 2 closed
- ✓ C. SW 1 open, SW 2 open

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# Chapter 7 – Sequential Switching Problems (18 of 22)

Here is the circuit used to determine the final value of the current in the second time segment. For this time segment, is the response form the natural response or the step response?




A. Natural response

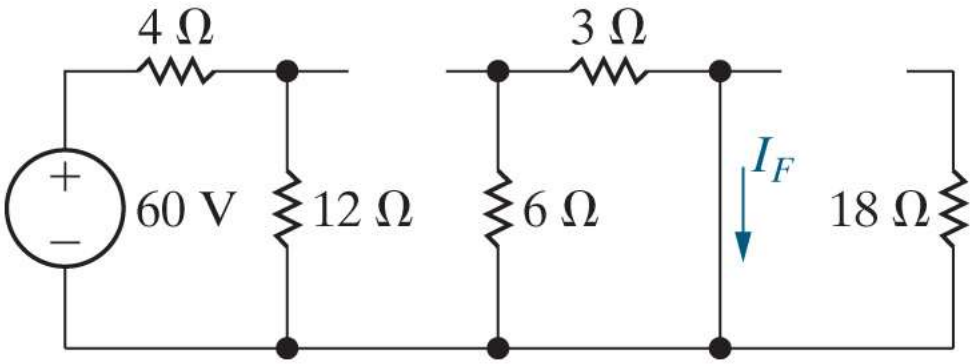
B. Step response

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# Chapter 7 – Sequential Switching Problems (18 of 22)

Here is the circuit used to determine the final value of the current in the second time segment. For this time segment, is the response form the natural response or the step response?



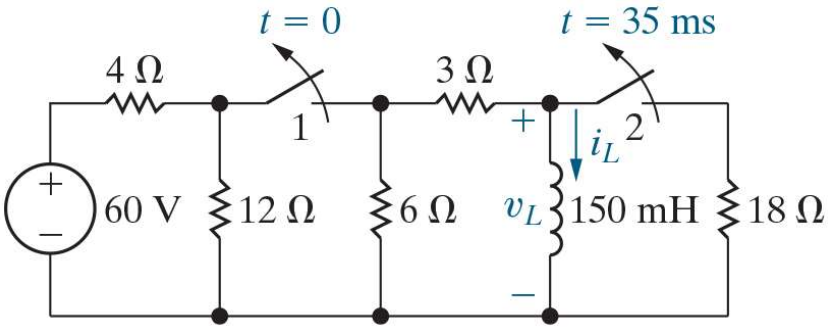

- ✔ A. Natural response
- ✘ B. Step response

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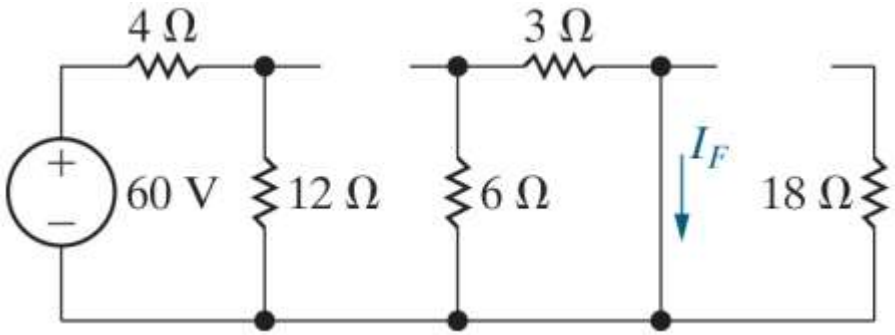
# Chapter 7 – Sequential Switching Problems (19 of 22)

Both switches have been closed for a long time. At t = 0, switch 1 is opened and stays open. Switch 2 opens 35 ms later and stays open. Find the expression for the voltage drop across the inductor for $t \geq 0^+$.




- Analysis for the second time segment, $t \geq 35$ ms:
  - Step 4 – Calculate the final value $I_f$.

$t \to \infty$:




Natural response!

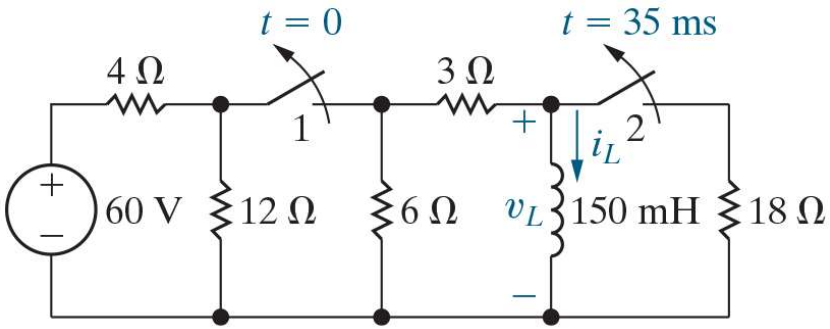
$\Rightarrow \quad I_F = 0$ A

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# Chapter 7 – Sequential Switching Problems (20 of 22)

Both switches have been closed for a long time. At t = 0, switch 1 is opened and stays open. Switch 2 opens 35 ms later and stays open. Find the expression for the voltage drop across the inductor for $t \geq 0^+$.




- Analysis for the second time segment, $t \geq 35$ ms:
  - Step 5 – Write the equation for $i_L(t)$

$$i_L(t) = I_f + (I_0 - I_f)e^{-[(t-t_0)/\tau]}$$

$$= 0 + (1.48 - 0)e^{-(t-0.035)/0.0167}$$

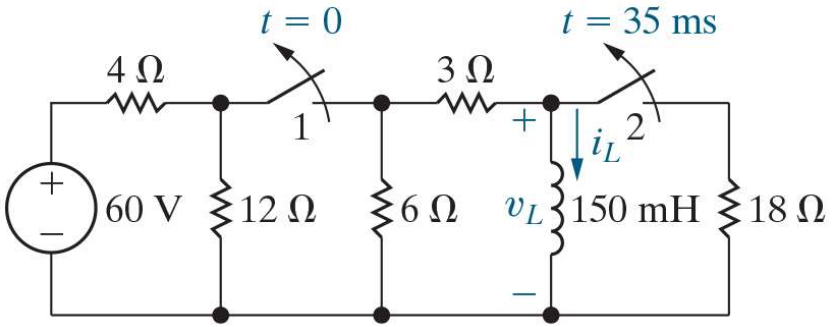
$$= 1.48e^{-60(t-0.035)} \text{ A}, \quad t \geq 35 \text{ ms}$$

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# Chapter 7 – Sequential Switching Problems (21 of 22)

Both switches have been closed for a long time. At t = 0, switch 1 is opened and stays open. Switch 2 opens 35 ms later and stays open. Find the expression for the voltage drop across the inductor for $t \geq 0^+$.




Summary:

$$i_L(t) = 6e^{-40t} \text{ A}, \quad 0 \leq t \leq 35 \text{ ms}$$

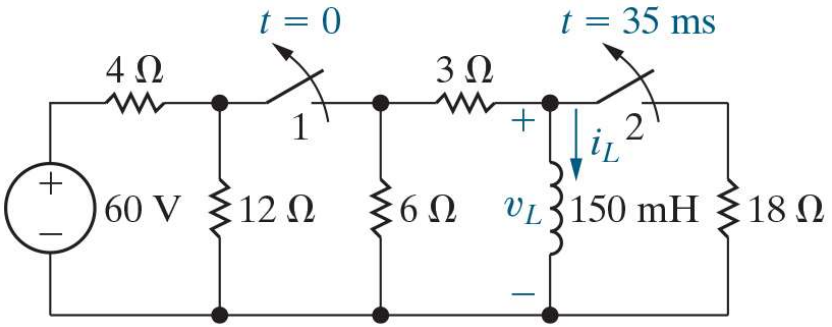
$$i_L(t) = 1.48e^{-60(t-0.035)} \text{ A}, \quad t \geq 35 \text{ ms}$$

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# Chapter 7 – Sequential Switching Problems (22 of 22)

Both switches have been closed for a long time. At t = 0, switch 1 is opened and stays open. Switch 2 opens 35 ms later and stays open. Find the expression for the voltage drop across the inductor for $t \geq 0^+$.




- Step 6 – Calculate other quantities of interest using $i_L(t)$.

$$v_L(t) = L\frac{di_L(t)}{dt}$$

$$v_L(t) = (0.15)(-40)(6e^{-40t}) = -36e^{-40t} \text{ V}, \quad 0 \leq t \leq 35 \text{ ms}$$

$$v_L(t) = (0.15)(-60)(1.48e^{-60(t-0.035)})$$

$$= -13.32e^{-60(t-0.035)} \text{ V}, \quad t \geq 35 \text{ ms}$$

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